

Curriculum-based teacher professional development in middle school science: A comparison of training focused on cognitive science principles versus content knowledge

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Funding information

Institute of Education Sciences, Grant/Award Number: R305C080009

Abstract

This study investigates, through a cluster-randomized trial, the effectiveness of two approaches to increasing middle school students' science learning using a traditional science curriculum. Ninety schools were randomly assigned into one of three arms: (a) a treatment arm in which the textbook curriculum was modified based on four principles of cognitive science coupled with teacher professional development (PD), (b) a second treatment arm in which teachers received PD designed to improve their knowledge of the science content, and (c) a business-as-usual control group. The PD was able to change teacher practice but barely improved teacher knowledge. No significant boost in student achievement was observed, except in a few instances, where there were some promising findings. Exploratory analyses were conducted to examine what makes the interventions more effective. Implications for future research were discussed.

KEYWORDS

cognitive science, curriculum design, middle school science, randomized controlled trial, student achievement, teacher content knowledge, teacher professional development

1 | INTRODUCTION

There is little doubt that preparing today's students for success in educational attainment, productive employment, informed citizenship, and overall well-being in a world of complex challenges requires dramatically upgrading the opportunities for all students to learn science. Though policy makers have long recognized this need (National Academy of Sciences, 2007; National Commission on Excellence in Education, 1983), domestic and international science assessments continue to demonstrate lackluster outcomes for US students. In the 2015 Program for International Student Assessment (PISA) test, the United States ranked 26th among countries, down two places from the 2012 test (Kelly, Nord, Jenkins, Chan, & Kastberg, 2013).

Recent initiatives such as the Next Generation Science Standards (NGSS Lead States, 2013), drawing on current research in the learning sciences, have outlined ambitious new standards to guide efforts to transform and improve K-12 science education in the United States (NGSS Lead States, 2013; National Research Council, 2007, 2012). NGSS calls for fundamental and comprehensive restructuring of science curricula, pedagogy, teacher professional development, and assessment. However, achieving those goals will require deploying and coordinating new strategies at multiple levels in educational systems.

Curriculum materials that support student engagement in science can make a difference in student learning (Songer, Kelcey, & Gotwals, 2009; Wilson, Taylor, Kowalski, & Carlson, 2010). Several studies examined the effectiveness of implementing inquiry-based or project-based curricula in science teaching through experimental studies and found mixed results (Borman, Gamoran, & Bowdon, 2008; Harris et al., 2015; Lynch, Pyke, & Grafton, 2012). Penuel, Gallagher, and Moorthy (2011) also conducted a randomized study to examine whether teacher professional development (PD) focused on giving teachers explicit instruction on teaching methods or guiding their selection of curriculum materials was effective. However, few studies have attempted to aggregate multiple well-established cognitive science principles and apply them simultaneously to existing science curricula, although cognitive science has proved to be an effective tool in promoting learning (National Research Council, 2000, 2007; Stein & Raudenbush, 2010). The current study, by improving existing curricula already in widespread use with research-based modifications that can be practically implemented under the current constraints in classrooms, fills a gap in both research and practice.

One key strategy is to incorporate findings from cognitive science research on learning, memory, and conceptual development into daily instruction in classrooms. We selected four principles to be implemented by teachers in standard classroom settings and integrated with each other to collectively address various phases and aspects of learning: (a) case comparisons; (b) visualization and diagrammatic reasoning; (c) spaced testing; and (d) conceptual challenges, misconceptions, and prior knowledge. Details on each of the principles are presented in the following section.

A traditional middle school science curriculum in widespread use, Holt Science and Technology (Holt), was modified based on the above four principles. The intervention targeted middle school because at that level science content becomes considerably more sophisticated, requiring students to understand and investigate explanatory systems that are no longer directly connected to the everyday world (Chi, Slotta, & De Leeuw, 1994). At the same time, middle school teachers often lack the disciplinary background to help their students grasp and reason about the more challenging concepts. The decision to focus on improving existing curricula rather than developing a new one was made because many districts have already invested in current curricula and lack resources to develop or adopt new materials and approaches.

Moreover, inadequate STEM content knowledge is often cited as an impediment to improving student outcomes, and teacher PD is an obvious candidate for additional investment (Chaney, 1995; Duschl, Schweingruber, & Shouse, 2007; Goldhaber & Brewer, 2000). Diamond, Maerten-Rivera, Rohrer, and Lee (2014) found that teacher content knowledge was the most significant teacher-level predictor of student science achievement. As both an active control for the cognitive science intervention (hereafter referred to as the Cog-sci group) and also as an intervention of interest in its own right, the study also utilizes a second treatment condition based on improving the science content knowledge of middle school teachers (the Content group). This study investigates both approaches and compares them to a business-as-usual control condition in a three-arm randomized control trial.

2 | THEORETICAL FRAMEWORK

Below we review literature on each principle and describe how they were incorporated in the curriculum modifications for three Holt units, each of which accounted for 2–3 months of instruction in the seventh or eighth grade. The three target units were Cells, Hereditary, and Classification (hereafter, Cells); Introduction to Matter (Matter); and Inside the Restless Earth (Earth). These units represent typical middle school content in biological science, physical science, and geology, respectively. We also review literature on the importance of teachers' content knowledge and pedagogical content knowledge.

2.1 | Cog-sci intervention

2.1.1 | Case comparisons

Teachers often draw on concrete examples or cases to illustrate concepts. Cognitive science research has found that a powerful way to use cases in learning is to simultaneously contrast two or more cases to draw out how they relate to a more abstract concept or process (Alfieri, Nokes-Malach, & Schunn, 2013; Schwartz & Bransford, 1998; Schwartz, Chase, Oppezzo, & Chin, 2011; Tang, Delgado, & Moje, 2014). Case comparison, which draws on reasoning processes such as analogical thinking and structural alignment (Gentner, 2010; Gentner & Holyoak, 1997; Gick & Holyoak, 1983), helps students see what is common across cases and what critical features differentiate similar concepts. Learning through case comparison transfers to tests and problem solving (Gick & Holyoak, 1983), and a number of studies have demonstrated that it can be used with traditional subject matter in classrooms (Gadgil & Nokes, 2009; Mason, 2004; Michael, Klee, Bransford, & Warren, 1993; Nagarajan & Hmelo-Silver, 2006; Star & Rittle-Johnson, 2009). Evidence shows that contrasting cases is more effective than studying multiple cases sequentially (Gentner, Loewenstein, & Thompson, 2003) and that case comparison is most effective when it is done before the target concepts are introduced through formal instruction rather than after instruction as a form of practice (Gentner, 2010; Schwartz et al., 2011). Presumably, the alignment, categorization, and reasoning processes that students engage in when they compare cases at the beginning of instruction help introduce the intellectual landscape for a new segment of study by attuning learners to the central characteristics of the relevant entities, processes, and phenomena.

One example of how case comparison was infused into the existing Holt materials is a series of activities over 2–3 class periods for the Matter unit. Working in small groups, students recorded their initial ideas about the properties of solids and liquids in a two-column table. Next, they added a few drops of food coloring to each sample and revised their lists based on new observations. Small

groups shared their observations in whole-class discussion, leading to a class list of the comparative properties of solids and liquids. The case comparison continued on a subsequent day with an investigation comparing liquids and gases. Overall, this sequence directed students' attention to key physical properties of solids, liquids, and gases that they would subsequently be able to explain when they learn that solids, liquids, and gases are forms of matter that are made up of particles and that how those particles move explains the distinctive properties of the different forms.

Each of the three Holt units was divided into several chapters, which were further divided into subsections, and a case comparison, which typically took one to two class periods, was included in most subsections. A typical unit included eight contrasting cases distributed over a calendar of 50 total instructional days, with approximately 15 of those days modified to include case comparison activities.

2.1.2 | Visualization and diagrammatic reasoning

Diagrammatic reasoning and the ability to interpret varied visual images that populate textbooks play a key role in science learning. The dissemination of scientific knowledge at all levels relies on charts, diagrams, photographs, and other visual representations, and learning depends heavily on accurate and thorough comprehension of various visualization conventions, such as the use of captions and labels, relative scale and magnification, arrows, color keys, and graphic perspectives (e.g., cutaways and zooms). Studies indicate that many students struggle to interpret graphs and diagrams correctly or ignore them altogether when reading a text (Canham & Hegarty, 2010; Chan & Unsworth, 2011; Florax & Ploetzner, 2010; Taylor & Hutton, 2013).

Drawing on research on instruction in comprehending visualizations both in classes and using technology (Schlag & Ploetzner, 2011; Schwaborn, Thillmann, Opfermann, & Leutner, 2011), we supplemented the curriculum by creating a series of 5- to 10-min visualization exercises interspersed throughout each unit. The exercises were generally led by the teacher, who used PowerPoint slides to present the examples—which were typically modified versions of graphics in the existing curriculum materials—and to guide the students in explicitly noticing and practicing graphical conventions and interpreting their meaning in context (e.g., when an arrow indicates a change of state, change over time, location, or one thing acting on another). The exercises grew in complexity over time to help students become fluent with diagrammatic conventions and thus process diagrammatic information more easily and also to help them improve their spatial thinking, which are all central to students' learning with scientific materials (Cromley et al., 2013, 2016; Schwonke, Berthold, & Renkl, 2009). For example, a visualization exercise toward the beginning of the Cells unit focused on how to interpret a diagram from the Holt text that uses a cutaway perspective, with the teacher using a cut orange to illustrate a view that exposes the inside of a three-dimensional object. Visualization exercises toward the end of the Cells unit involved coordinating multiple elements, such as labels and captions, color, zoom perspectives, and arrows, to interpret diagrams of photosynthesis and cellular respiration. Students also practiced writing their own captions for complex diagrams. Typically, one-third of the total instructional days included a short visualization or diagrammatic reasoning activity.

2.1.3 | Spaced testing

One difficulty for both teachers and students is that students often quickly forget what they have been taught. Cognitive science has shown that the durability of learning can be dramatically increased by periodically revisiting material that was learned earlier, a phenomenon known as the *spacing effect*

(Pashler et al., 2007; Rawson & Dunlosky, 2011; Rohrer & Pashler, 2007). A similarly powerful technique for improving retention takes advantage of the so-called *testing effect* (Roediger & Karpicke, 2006); it involves having students actively retrieve material from long-term memory as opposed to hearing a lecture, passively reviewing notes or texts, or repeating material that is already active in working memory. Unlike high-stakes tests, the testing effect involves frequent low-stakes retrieval practice during learning (Dunlosky, Rawson, Marsh, Nathan, & Willingham, 2013; Metcalfe & Kornell, 2007). The curriculum modifications combined spaced practice and active testing in daily warm-up questions and end-of-section quizzes. Daily warm-up questions prompted students to explain a previous day's main ideas to recapitulate the purpose and content of the lesson. Weekly end-of-section quizzes, which included questions from both the current and previous sections of the unit, were short and easy to grade by teachers, and time was scheduled to allow for review of quizzes on a subsequent day. The modifications followed a map to ensure that all main ideas were revisited on multiple occasions over the 2–3 months that it took to complete each unit.

2.1.4 | Conceptual challenges, misconceptions, and prior knowledge

Cognitive research has shown that one of the strongest predictors of how well a student learns something new is how relevant prior knowledge is organized by the student (Ausubel, 1968; National Research Council, 2000). However, in science, middle school students often lack relevant conceptual frameworks to support new learning. If students cannot relate new information to a meaningful framework, they typically resort to memorizing terms that will not be connected or applied to other relevant knowledge—and thus quickly forgotten. In the extreme, intuitive frameworks compounded over years may take the form of active misconceptions and interfere with new instruction (Chi, 2005; Vosniadou, Skopeliti, & Ikospentaki, 2004).

Fostering deep conceptual change, connecting prior knowledge to more sophisticated scientific frameworks, and countering misconceptions is a long, gradual process (Carey, 1986; National Research Council, 2007, 2012) that often entails constructing new ontologies with students (Chi et al., 1994) and building new explanatory systems to accommodate their enhanced understanding of the phenomena (Chi, 2005, 2008; Smith, Maclin, Grosslight, & Davis, 1997). The constraint of maintaining the structure and timeline of the existing Holt curriculum precluded the kinds of reorganization, expansion, and change of focus characteristic of new curricula that are specifically designed for conceptual change or for extended learning progressions in science (Catley, Lehrer, & Reiser, 2004; Corcoran, Mosher, & Rogat, 2009; Duncan, Castro-Faix, & Choi, 2016; Duncan, Rogat, & Yarden, 2009; Duschl et al., 2007; Stevens, Delgado, & Krajcik, 2010; Talanquer, 2009; Wiser, Smith, & Doubler, 2012). As a result, unlike in the case of the other three principles, which were used as the basis for newly devised stand-alone activities throughout the units, we embedded this fourth principle in modifications related to the other three to target challenging concepts. The three other Cog-sci modifications were used as much as possible to target concepts known to be conceptually difficult for learners. Contrasting cases were deployed to help students construct a productive conceptual framework into which new ideas could be assimilated. Visualization and diagrammatic reasoning activities provided students with tools to understand images and representations that illustrate abstract, dynamic processes that cannot be directly observed. Fluency with these scientific representations can help students reason with the conceptual systems that go beyond everyday experience. Spaced practice and testing ensured that important concepts were systematically revisited throughout the unit and activated and connected to new topics.

We intended to facilitate learning by evoking background knowledge in a way that encourages students to process, revise, and connect new concepts to prior knowledge, thereby integrating scaffolds for more developed explanatory systems and countering misconceptions (Chinn & Malhotra, 2002; Mason, 2001; Vosniadou, Ioannides, Dimitrakopoulou, & Papademetriou, 2001).

The Matter unit provides one example of how we selected and developed modifications to support the learning of a particle model of matter (Smith, Wiser, Anderson, & Krajcik, 2006; Talanquer, 2009; Wiser et al., 2012). Several contrasting cases were introduced to help students see how the macroscopic properties of matter are related to a particle model explanation. Subsequent contrasting cases provided opportunities to compare different ways of representing particle models and to differentiate between mass and weight, allowing students to see what differentiates them and how they were related. Visualization activities and daily warm-ups regularly revisited the ideas that all matter has mass and takes up space, and they provided repeated opportunities for students to work with the critical concepts of weight, density, and volume.

The Cog-sci intervention targeted teachers pedagogical content knowledge, or the specialized knowledge necessary to convey content effectively to students (Shulman, 1986). Teachers received explicit instruction and materials on how to integrate the cognitive science principles into their classroom instruction, including a detailed Cognitive Science Manual (CSM), prepared by our research team that provided guidelines, lessons, warm-ups, and assessments for each of the three intervention units. This approach is consistent with the considerable literature indicating that a deep understanding of principles of learning, combined with specific strategies to teach those principles, can foster student understanding (Ball, 1991, 2000).

2.2 | Content intervention

Another common approach to teacher learning in science is to focus on content. There is a general acceptance of the notion that general content knowledge in a subject like science, such as might be imparted via a university textbook, is distinct from pedagogical content knowledge, which incorporates knowledge of how to model and illustrate concepts and how students learn in a particular content area. Federal and state governments invest billions of dollars annually in PD programs designed to improve teachers' subject-specific content knowledge (Birman et al., 2009; U.S. Department of Education, 2014). Although there is evidence that having teachers with higher levels of subject matter knowledge is beneficial for students (Chaney, 1995; Goldhaber & Brewer, 2000) and that changing teachers through content-focused PD is essential for improving student learning (Fishman, Marx, Best, & Tal, 2003), the optimal level of content knowledge and how to achieve that for teachers is not well understood, especially for elementary and middle grade teachers (Duschl et al., 2007). While hundreds of studies have addressed the topic of teacher learning and PD (Clewell, Campbell, & Perlman, 2004; Garet 2016, Porter, Desimone, Birman, & Yoon, 2001; Supovitz, 2001; Yoon, Duncan, Lee, Scarloss, & Shapley, 2007), results are generally mixed due to the large variation in design, implementation, the target populations, and target outcomes (Desimone & Garet, 2015; Kennedy, 2016; van Driel, Meirink, van Veen, & Zwart, 2012; Yoon et al., 2007).

Motivated by the belief that to be an effective teacher, one needs to possess the relevant content knowledge (Garet et al., 2001; Grossman, 1990), this study incorporated a second active intervention condition—the Content intervention—which focused on enhancing teachers' content knowledge related to the three targeted Holt science units, with the hope that increased teacher knowledge would transfer to improved student learning. The goal of the Content intervention was to improve teachers' understanding of the science subject matter as adult learners, and the content was pitched at a level

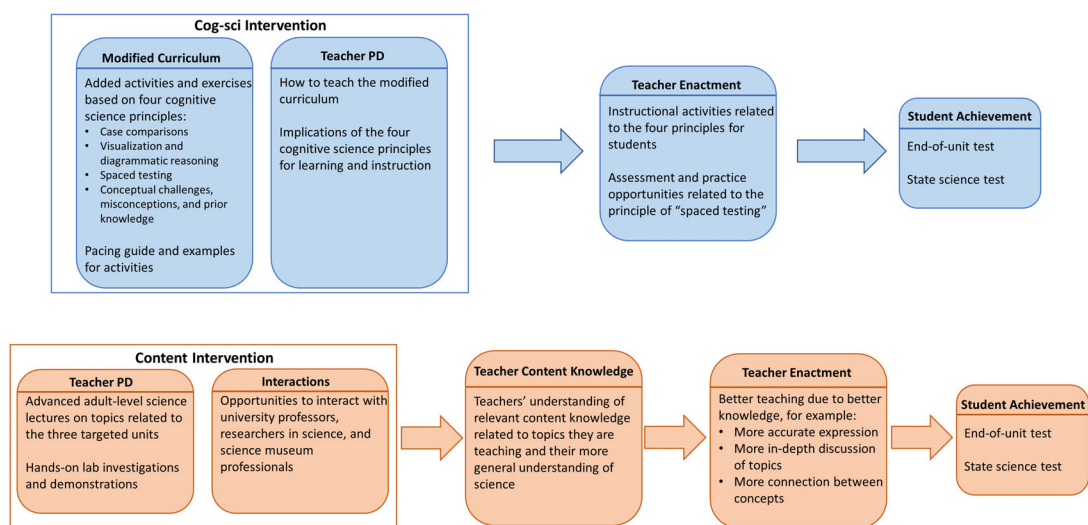


FIGURE 1 Theoretical framework [Color figure can be viewed at wileyonlinelibrary.com]

similar to an introductory undergraduate science course. For example, teachers in the Content arm for the Matter unit explored deeper explanations of states of matter and changes of state by studying different forms of particle movement and molecular bonds. They also worked with more advanced mathematical representations, such as graphical representations of solubility curves for compounds in water, and the formulas for gas laws relating pressure, volume, temperature, and amount.

Many PD programs lack a clear theory that connects teacher training and student learning outcomes (Gore et al., 2017), and it has been referred to as “evidence void” when the effects on teacher knowledge or practices are missing (Kennedy, 2016). Our study tried to test the complete chain of causality from PD to student achievement and the mediating variables in between. Figure 1 presents the theoretical framework on how the two interventions are hypothesized to improve student learning. The causal assumptions of the Cog-sci intervention are illustrated in blue, while the assumptions of the Content intervention are illustrated in orange.

3 | DESIGN AND DATA COLLECTION

3.1 | Design

The study was carried out in a large urban district in the northeast region of the country. The district has more than 200 public schools with more than 100,000 students enrolled. More than 90% of the students are eligible for free or reduced-price lunch. Teachers from 90 schools participated in the study. To promote teacher collaboration and minimize contamination across conditions, a cluster-randomized trial was conducted—schools were randomized into one of the three arms: The control arm, the Content arm, or the Cog-sci arm. School locations were treated as blocks to obtain a balanced number of intervention schools across the different regions in the school district. The design included two cohorts of students, with each cohort unfolding across seventh and eighth grades.

To reduce confounding, we designed both the content-only and the cognitive science PD to have the same structure, duration, and core features despite the Content arm's focus on content only and the Cog-sci arm's focus on pedagogy and curricular materials. Both followed the tenets of high-quality professional development as set forth in the literature: They focused on subject-area content,

alignment with the curriculum and school priorities, provision of active learning opportunities, and collaborative engagement, and they occurred over a substantial span of time (e.g., Desimone, 2009; Kennedy, 2016). Additionally, the PD accounted for potential organizational constraints (van Driel et al., 2012) by syncing the timing of the PD and its lessons to the cadence of curriculum pacing and working with the districts to ensure teachers had sufficient time to participate.

For both arms, we provided a 2.5-day (18-hr) summer PD session and four 2-hr professional learning community (PLC) sessions during the school year for 2 years, for a total of 34 hr of PD. The PD for both intervention arms happened in the summer and the following school year. Teacher content knowledge was assessed after the summer PD sessions. Ongoing sessions during the school year, which were called professional learning community meetings (PLCs), took place approximately monthly during the period when the teachers were teaching a given target unit. The Cells unit was taught in the fall of seventh grade, and the Matter and Earth units were taught in the fall and spring of eighth grade, respectively. In the first summer of the study, only Cells PD was held. In the second summer, Cells PD was held for only new teachers in the study, and Matter and Earth PDs were held for all eighth-grade teachers. In the third summer, Matter and Earth PD sessions were held for only new teachers joining the study. Any teacher who participated in both cohorts of a given unit received a total of 26 hr of PD and PLC in the first year and 8 hr of PLC in the second year, adding up to 34 hr in a 2-year span. A teacher who participated in two units would receive a maximum of 68 hr of PD across 2 years.

In seventh grade, students in participating schools received Cells instruction and took the Cells end-of-unit test in the fall. In eighth grade, students received Matter instruction and took the Matter end-of-unit test in the fall. They participated in the Earth instruction and were tested the following spring. At the end of eighth grade, they also took the state science test. The first cohort entered seventh grade in the fall of 2009 and finished eighth grade in the spring of 2011. The second cohort entered seventh grade in the fall of 2010 and finished eighth grade in the spring of 2012. See Figure 2 for the study timeline. Time is represented on the top bar, and the two arrows represent the two cohorts of students.

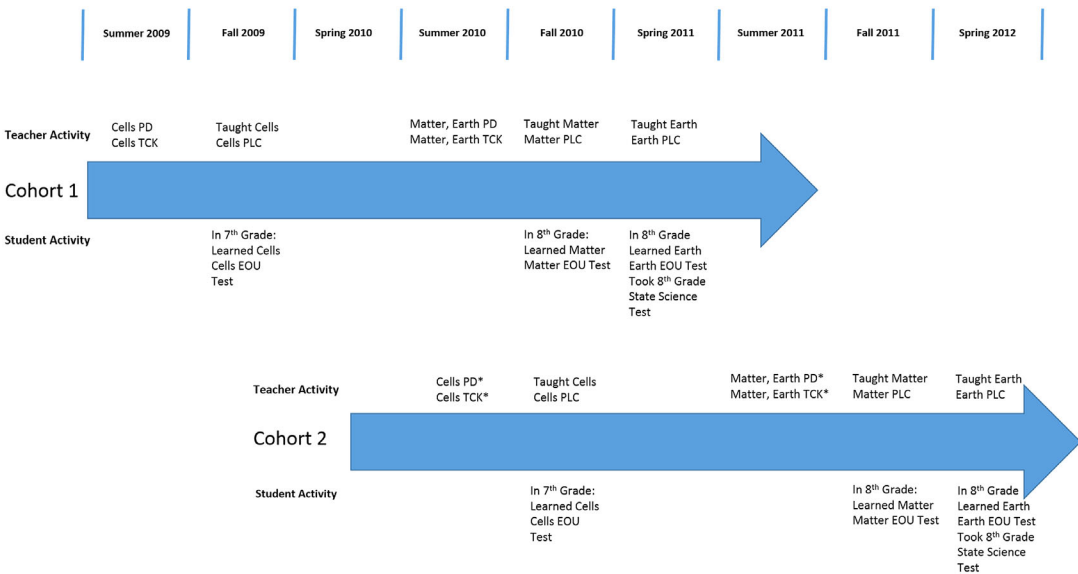


FIGURE 2 Intervention timeline. PD, professional development; PLC, professional learning community; TCK, teacher content knowledge test; EOU, end-of-unit test. *Activities for teachers that were new to the study and did not participate in the PD activity the previous year [Color figure can be viewed at wileyonlinelibrary.com]

Power analyses were conducted prior to the recruitment of schools using PowerUp! (Dong & Maynard, 2013). Assuming a minimum detectable effect size of 0.2, the percentage of variance at the student level to be 80%, the percentage of variance explained at the student level to be 50%, and the percentage of variance explained at the higher levels to be 60% (Klein, Hamilton, McCaffrey, & Stecher, 2000), we would need 85 schools to have a power of 0.8. A total of 90 schools agreed to participate in the study, and random assigned were conducted for those 90 schools.

3.2 | Data

For Cohort 1, the final analysis file contained 90 schools, 145 teachers, and 6,410 students. For Cohort 2, the final analysis file contained 82 schools, 130 teachers, and 5,508 students. Schools were randomly assigned to the three arms of the experiment, and in each school all seventh- and eighth-grade science teachers participated in the assigned treatment or control condition for that school. Table 1 summarizes the descriptive statistics for participating teachers.

Despite random assignment of schools, Cog-sci group teachers were significantly less experienced than teachers in the other two groups. There were also significantly more White teachers in the Cog-sci group than in the other two groups in Cohort 1.

Out of the 40 Cohort 2 teachers in the control group, 30 were returning teachers who also taught Cohort 1 students. Thirty-two out of 41 teachers were returning teachers for the Content group, and 29 out of 43 teachers were returning teachers for the Cog-sci group. The teacher attrition rate between Cohort 1 and Cohort 2 was 40% for the control group, 30.4% for the Content group, and 42% for the Cog-sci group. None of the pairwise differences were statistically significant at the 0.05 level. We followed up with teachers who did not return to teach in Cohort 2. The top two causes for attrition (which accounted for more than 40% of all attrition) were “assigned to teach another grade” or “assigned to teach another subject,” neither of which were likely related to low student

TABLE 1 Descriptive statistics for teachers, by cohort and treatment condition

Treatment condition	Cohort 1			Cohort 2		
	Control	Content	Cog-sci	Control	Content	Cog-sci
Number of observations	54	42	49	47	38	37
Proportion female	0.8 (0.4)	0.9* (0.4)	0.7 (0.5)	0.7 (0.5)	0.8 (0.4)	0.8 (0.4)
Age	44.5 (9.1)	44.6 (11.0)	37.9 (13.1)	43.8 (8.8)	43.3 (12.0)	38.0 (14.2)
Proportion white	0.6 (0.5)	0.5* (0.5)	0.8 (0.4)	0.7 (0.5)	0.6 (0.5)	0.8 (0.4)
Years of teaching	14.1* (9.8)	16.1* (11.2)	7.5 (7.0)	14.4* (9.3)	14.3* (10.4)	6.8 (5.6)
Percentage with master's degree	0.04 (0.2)	0.00 (0.0)	0.08 (0.3)	0.00 (0.0)	0.00 (0.0)	0.06 (0.2)

Note: Standard deviations are in parenthesis. Proportion tests and *t*-tests were conducted to examine the statistical significance of all pairwise comparisons. No significant differences were found for the comparison between the control group and the Content group. An asterisk represents a significant difference between that group and the Cog-sci group.

**p* < .05.

performance. We also examined the relationship between teachers' propensity for attrition and student achievement and did not find the relationship to differ by arm.¹ Given these findings, the differential attrition rate should not be worrisome.

Table 2 summarizes the participating students' background information by condition and cohort. For all background characteristics, we calculated Hedges' g , and the effect sizes were all less than 0.25 of the pooled standard deviation, which meets What Works Clearinghouse's standards (Seftor

TABLE 2 Descriptive statistics for students, by cohort and treatment condition

Treatment condition	Cohort 1			Cohort 2		
	Control	Content	Cog-sci	Control	Content	Cog-sci
Number of observations	2,267	1,960	2,183	1,852	1,826	1,830
Age in months	164.9 (6.5)	164.8 (6.4)	164.4 (7.3)	152.5 (6.0)	152.3 (6.3)	152.3 (6.2)
Proportion female	0.5 (0.5)	0.5 (0.5)	0.5 (0.5)	0.5 (0.5)	0.5 (0.5)	0.5 (0.5)
Proportion non-Hispanic white	0.2 (0.4)	0.1 (0.3)	0.1 (0.3)	0.2 (0.4)	0.1 (0.3)	0.1 (0.3)
Proportion African American	0.5 (0.5)	0.6 (0.5)	0.6 (0.5)	0.6 (0.5)	0.6 (0.5)	0.6 (0.5)
Proportion Hispanic	0.2 (0.4)	0.2 (0.4)	0.2 (0.4)	0.2 (0.4)	0.2 (0.4)	0.2 (0.4)
Proportion Asian	0.1 (0.3)	0.1 (0.2)	0.1 (0.3)	0.1 (0.3)	0.1 (0.3)	0.1 (0.2)
Proportion other races	0.0 (0.1)	0.0 (0.1)	0.0 (0.1)	0.0 (0.1)	0.0 (0.1)	0.0 (0.1)
Proportion eligible for free/reduced-price lunch	0.8 (0.4)	0.9 (0.3)	0.8 (0.4)	0.7 (0.5)	0.8 (0.4)	0.7 (0.5)
Proportion identified as English language learners	0.2 (0.4)	0.1 (0.3)	0.2 (0.4)	0.1 (0.2)	0.1 (0.2)	0.0 (0.2)
Proportion identified as special education students	0.2 (0.4)	0.2 (0.4)	0.1 (0.3)	0.1 (0.4)	0.2 (0.4)	0.1 (0.3)
Fifth-grade math test score	1,323.7 (213.1)	1,307.8 (219.7)	1,342.3 (218.9)	1,346.3 (205.6)	1,329.2 (228.5)	1,348.1 (212.1)
Sixth-grade math test score	1,362.8 (222.3)	1,325.2 (217.9)	1,349.0 (221.1)	1,381.8 (219.5)	1,361.7 (234.0)	1,361.4 (209.1)
Fifth-grade Reading test score	1,195.1 (217.5)	1,172.6 (208.8)	1,219.1 (218.7)	1,216.4 (211.8)	1,189.9 (221.0)	1,227.9 (219.9)
Sixth-grade Reading test score	1,252.6 (214.3)	1,221.3 (203.1)	1,257.1 (213.2)	1,271.3 (213.2)	1,240.2 (220.5)	1,257.7 (210.9)

Note: Standard deviations are in parenthesis. Differences between groups were calculated, and none of the differences were larger than 0.25 of a standard deviation. ANOVAs were conducted for all student characteristics, and none were significant.

et al., 2011). We decided to include all those background characteristics in the analytic models to adjust for their differences.

4 | OBJECTIVES AND METHODS

4.1 | Research questions

The main goal of the study was to test the effects of the two interventions on student achievement. Based on our theoretical framework, effects on students would channel through teachers' improved knowledge or practice, and thus we tested whether or not the intervention impacted teacher knowledge or practice. Before conducting the study, we hypothesized that the Cog-sci intervention would work better for Cohort 2 because as Cog-sci teachers gained experience with cognitive science principles, they would be able to teach the unit more effectively. Granted that teacher attrition was high and that the Cohort 2 impact estimates might not be able to truly reflect a "second-time teaching" effect, we still wanted to explore if returning teachers in the Cog-sci group help more with student learning.

Our research questions (RQs) were as follows:

1. What are the impacts of the Cog-sci intervention and the Content intervention on teacher knowledge and teacher practice?
2. What are the impacts of the Cog-sci and Content interventions on student achievement?
3. Do prior experiences receiving the intervention help Cog-sci teachers more in promoting student learning for Cohort 2?

4.2 | Measures

Creation of the measures was guided by our theoretical framework. The measures are sorted into the following categories based on their relevance to the research questions.

4.2.1 | Outcome measures for teachers

For each of the three units, a teacher knowledge test (TKT) was constructed. Using a pool of items drawn from multiple published tests, the authors developed an automated test construction algorithm to pick items so that the final test was aligned to the unit based on the Survey of Enacted Curriculum (SEC). The SEC measures content in a two-dimensional framework of content by cognitive demands (Council of Chief State School Officers, 2000; Porter, 2002), and it has been used by researchers, states, districts, schools, and teachers nationwide to analyze alignment among instruction, curriculum materials, standards, and assessments (Porter, McMaken, Hwang, & Yang, 2011; for details of the test construction procedure, see Porter, Polikoff, Barghaus, & Yang, 2013). Items in the pool were drawn from PRAXIS, Diagnostic Science Assessments for Middle School Teachers, and MOSART. The total number of items for the Cells, Matter, and Earth tests were 20, 25, and 25, respectively, and the Cronbach's alphas were 0.71, 0.76, and 0.90. A one-parameter IRT model was fit to the items to examine item fit and item difficulty and to calculate ability scores.² The range of item difficulties provided good coverage for the range of teacher abilities measured. All items had good fit statistics (both the infit and the outfit statistics were in the range of 0.7–1.2). Because the raw total score and the IRT score had a correlation of 0.99 and their distributions were similar, we decided to use the raw total scores since they were easier to interpret.

The research team also created items in the teacher implementation survey to measure to what degree a teacher used the first three cognitive science principles. The fourth principle was embedded in modifications related to other three and did not include stand-alone activities in any of the three arms. Questions about use of cognitive science-related practices were phrased to be interpretable and meaningful to teachers in any arm of the study. Three scales (one for case comparisons, one for visualization, one for spaced testing) were constructed. The scales have 8, 9, and 7 items, respectively, and the Cronbach's alphas are 0.88, 0.82, and 0.73. One-parameter partial-credit models were fit to items within each of the three scales. All items had good fit statistics, and there were no reversed threshold parameter estimates. The correlations between the three scale scores were only moderate (from 0.26 to 0.41), and we decided not to calculate composite scores for implementation. The correlations between the raw average score and the IRT score were very high. We decided to use the IRT scores when predicting teacher practice because the distribution of those scores was closer to a normal distribution (available in Figure S1). Details about the three scales, the items, and the rating categories are available in Tables S1–S3.

4.2.2 | Outcome measures for students

The first type of student outcome measure was the end-of-unit test. The authors applied the same procedure as described above in the teacher outcome section. The total number of items for each of the three tests was 18 (items came from the eighth-grade science assessment in the following states: Florida, Illinois, Indiana, Maine, Massachusetts, Michigan, and Ohio). They had moderate to high alignment with the curriculum, and the Cronbach's alphas were 0.72, 0.74, and 0.76, indicating satisfactory reliability. A one-parameter IRT model was fit to the items to examine item fit (both the infit and the outfit statistics were in the range of 0.7–1.2) and item difficulty and to calculate ability scores. The correlations between the raw scores and the IRT scores were 0.99, and their distributions were similar. The authors decided to use raw scores in the analyses since they were easier to interpret.

The second type of student outcome measure was the state science test given at the end of eighth grade. The total raw score for the state test was 65, with an alpha of 0.92. Because the state test was administered in a formal setting and the stakes were relatively higher, students may have been more likely to take it seriously compared with the end-of-unit tests. The downside of using the state test was that the content covered by the test might misalign with the target units. To address misalignment, we selected items from the state test to construct subtests based on the same automated test construction algorithm (Porter et al., 2013). Most subtests were reliable and had moderate alignment with the target curriculum.³ IRT models were fit to examine item fit and calculate ability estimates. In the end, we used raw scores because there was little difference between the two types of scores and the raw scores were easier to interpret.

4.2.3 | Treatment indicators

Three dummy indicators were created to represent the Cog-sci group, the Content group, and the control group. Depending on the type of model, one of the three groups was treated as the reference group.

4.2.4 | Covariates

Teacher gender, ethnicity, degree attainment, years of teaching, and age were used as covariates in models predicting teacher outcomes. Student gender, ethnicity, limited English proficiency (LEP)

TABLE 3 Coefficient estimates from HLMs, teacher knowledge test scores as response

	Teacher knowledge test		
	Cells	Matter	Earth
	Mean = 13.6, <i>SD</i> = 3.2	Mean = 18.2, <i>SD</i> = 3.9	Mean = 16.7, <i>SD</i> = 4.1
Female	−1.15 (0.87)	−1.93 (1)	−0.53 (0.98)
Age	0.01 (0.04)	0.03 (0.05)	−0.02 (0.05)
White	0.6 (0.83)	1.67 (1)	1.59 (1.01)
Major in STEM	−0.72 (0.87)	−0.37 (1.07)	−0.48 (1.05)
Master's degree or above	0.41 (2.21)	1.7 (2.95)	−1.16 (2.86)
Total years of teaching	−0.02 (0.05)	−0.02 (0.06)	0.02 (0.06)
Cog-sci group indicator	−0.32 (0.97)	−0.71 (1.18)	−2.16 (1.2)
Control group indicator	−0.93 (0.97)	−2.1 (1.21)	−3.17* (1.19)
Intercept	14.46 (1.78)	16.58 (2.21)	19.93 (2.18)

Note: Under the unit names are the means and standard deviations for the response variable in the model. The Content group was treated as the reference group.
* $p < .05$.

status, disability status, free or reduced-price lunch (FRL) status, and state test scores for fifth and sixth grade mathematics and reading were used as student-level covariates. The means of those variables were also calculated at the classroom and school levels and used as classroom- and school-level covariates.

4.3 | Methods

4.3.1 | RQ 1

Two-level hierarchical linear models (HLMs; teacher level and school level) were used to model the impact of the interventions on teacher knowledge and teacher implementation. Because teachers took the teacher knowledge test only once for each unit no matter how many cohorts they taught, we combined Cohort 1 and Cohort 2 teachers. Because teachers filled out a survey every time they finished teaching a unit, we fit HLMs for each unit and cohort combination to compare teacher practice measures (Rethinam, Pyke, & Lynch, 2008).

4.3.2 | RQ 2 and RQ 3

Three-level HLMs (student level, classroom level, and school level) were used when modeling student end-of-unit test scores. Teachers were not treated as an additional level between classrooms and schools because it was hard to separate variance at the teacher level from that at the school level.⁴ Teacher-level characteristics were included at the classroom level in the models. For impact models on state test and subtests, we used a two-level HLM (student level and school level). Details of the HLMs are available in supplemental materials.

TABLE 4 Coefficient estimates from HLMs, teacher implementation scores as response

	Implementation of case comparisons						Implementation of spaced testing						Implementation of visualization and diagrammatic reasoning					
	Mean = 0.49, standard deviation = 2.09						Mean = 1.1, standard deviation = 1.07						Mean = 0.38, standard deviation = 0.72					
	Cells	Matter	Earth	Cohort 1	Cohort 2	Cohort 1	Cells	Matter	Earth	Cohort 1	Cohort 2	Cohort 1	Cells	Matter	Earth			
	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2			
Female	0.71 (0.67)	-0.66 (0.57)	-0.08 (0.54)	-0.63 (0.45)	-0.62 (0.51)	-0.15 (0.31)	-0.15 (0.28)	0.22 (0.27)	-0.52 (0.30)	-0.16 (0.36)	0.49* (0.21)	0.32 (0.19)	0.17 (0.13)	0.3 (0.18)	-0.07 (0.22)			
Age	0.01 (0.03)	0.03 (0.03)	0.02 (0.03)	0.02 (0.03)	-0.03 (0.03)	0.03 (0.01)	0.02 (0.02)	0.02 (0.02)	0.01 (0.02)	-0.01 (0.02)	-0.01 (0.01)	-0.01 (0.01)	0 (0.01)	-0.02 (0.01)	0.02 (0.01)			
White	-0.89 (0.62)	0.41 (0.59)	-0.42 (0.59)	0.7 (0.55)	-0.14 (0.52)	-0.02 (0.29)	0.55 (0.29)	0.16 (0.30)	0.29 (0.34)	-0.32 (0.36)	0.21 (0.20)	0.17 (0.21)	0.36* (0.15)	0.46* (0.23)	0.51* (0.22)			
Major in STEM	-0.67 (0.63)	-1.22 (0.61)	-0.13 (0.60)	0.18 (0.60)	0.67 (0.55)	-0.11 (0.29)	0 (0.30)	0.45 (0.30)	-0.39 (0.32)	0.09 (0.38)	-0.19 (0.20)	0.04 (0.22)	0.11 (0.15)	-0.12 (0.32)	0.15 (0.23)			
Master's degree or above	4.75* (1.55)	4.86*** (1.47)	2.59 (2.21)	4.19*** (1.32)	1.69 (1.85)	1.82* (0.72)	1.68* (0.73)	0.82 (1.10)	1.62* (0.73)	0.04 (1.30)	1.92*** (0.49)	1.21* (0.52)	0.38 (0.54)	0.64 (0.70)	0.59 (0.80)			
Total years of teaching	0.03 (0.04)	-0.07 (0.03)	0 (0.04)	0 (0.03)	0.05 (0.03)	-0.02 (0.02)	-0.02 (0.02)	0 (0.02)	-0.02 (0.02)	-0.01 (0.02)	0.01 (0.01)	-0.01 (0.01)	0.03** (0.01)	0.01 (0.02)	0.01 (0.01)			
Content group indicator	-2.45*** (0.73)	-1.96* (0.68)	-2.72*** (0.72)	-0.81*** (0.73)	-1.36 (0.65)	-0.43 (0.34)	-0.1 (0.34)	-0.46 (0.36)	0.03 (0.39)	-1 (0.46)	-0.51* (0.23)	-0.54* (0.24)	-0.2 (0.18)	-0.14 (0.38)	-0.02 (0.28)			
Control group indicator	-1.69* (0.67)	-1.06 (0.62)	-1.53* (0.65)	-0.13 (0.63)	-1.01 (0.54)	-0.18 (0.31)	-0.03 (0.31)	-0.74* (0.32)	0.14 (0.34)	-0.35 (0.39)	-0.15 (0.21)	-0.13 (0.23)	-0.18 (0.16)	-0.42 (0.34)	0.07 (0.24)			
Intercept	1.19 (1.27)	1.23 (1.23)	1.17 (1.32)	-0.64 (1.13)	2.27 (1.21)	0.42 (0.58)	0.46 (0.61)	-0.11 (0.66)	0.8 (0.68)	2.61 (0.82)	0.31 (0.40)	0.54 (0.43)	-0.23 (0.32)	0.84 (0.50)	-0.77 (0.51)			

Note: Under the unit names are the means and standard deviations for the response variable in the model. The Cog-sci group was treated as the reference group.
* $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 5 Coefficients estimates from HLMs, student end-of-unit test scores as response

	Cells		Matter		Earth	
	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2
<i>School-level predictors</i>						
Content intervention indicator	−0.42 (0.38)	−0.17 (0.48)	0.17 (0.42)	−0.69 (0.52)	−0.25 (0.51)	−0.27 (0.60)
Cog-sci intervention indicator	0.22 (0.38)	0.17 (0.46)	0.76 (0.39)	0.71 (0.48)	−0.01 (0.47)	0.67 (0.55)
Centered school average 5th grade math score	0.005 (0.01)	−0.006 (0.01)	0.002 (0.01)	0.001 (0.01)	−0.012 (0.01)	−0.007 (0.01)
Centered school average sixth grade math score	0.005 (0.00)	−0.003 (0.01)	0.007 (0.01)	0 (0.01)	0.01 (0.01)	0.002 (0.01)
Centered school average fifth grade Reading score	−0.002 (0.01)	0.014* (0.01)	−0.006 (0.01)	0.004 (0.01)	0.018* (0.01)	0.003 (0.01)
Centered school average sixth grade Reading score	0.003 (0.01)	0.008 (0.01)	0.01 (0.01)	0.006 (0.01)	−0.002 (0.01)	0.012 (0.01)
<i>Classroom-level predictors</i>						
Centered classroom average 5th grade math score	0.003 (0.00)	0.003 (0.00)	0.005 (0.00)	0.001 (0.00)	0.005 (0.00)	0.004 (0.00)
Centered classroom average sixth grade math score	0.003 (0.00)	0.004 (0.00)	0.006 (0.00)	0.009** (0.00)	0.005 (0.00)	0.002 (0.00)
Centered classroom average 5th grade Reading score	0.004 (0.00)	0.003 (0.00)	0.003 (0.00)	0.005 (0.00)	0.006 (0.00)	0.003 (0.00)
Centered classroom average sixth grade Reading score	0.001 (0.00)	0.002 (0.00)	−0.001 (0.00)	0 (0.00)	−0.002 (0.00)	0.006 (0.00)
<i>Student-level predictors</i>						
Centered student fifth grade math score	0.001*** (0.00)	0.002*** (0.00)	0.003*** (0.00)	0.002*** (0.00)	0.002*** (0.00)	0.001 (0.00)
Centered student sixth grade math score	0.002*** (0.00)	0.002*** (0.00)	0.002*** (0.00)	0.003*** (0.00)	0.001** (0.00)	0.003*** (0.00)
Centered student fifth grade Reading score	0.002*** (0.00)	0.002*** (0.00)	0.001** (0.00)	0.002*** (0.00)	0.002*** (0.00)	0.003*** (0.00)
Centered student sixth grade Reading score	0.003*** (0.00)	0.003*** (0.00)	0.003*** (0.00)	0.003*** (0.00)	0.005*** (0.00)	0.003*** (0.00)
Female	−0.32*** (0.08)	−0.29** (0.09)	−0.22* (0.09)	−0.38*** (0.09)	−0.08 (0.10)	−0.59*** (0.10)
Black	−0.17 (0.15)	−0.42* (0.17)	−0.57*** (0.16)	−0.6*** (0.16)	−0.38* (0.18)	−0.51** (0.18)
Hispanic	0.03 (0.17)	−0.19 (0.19)	−0.5** (0.18)	−0.32 (0.17)	−0.3 (0.20)	−0.19 (0.20)

(Continues)

TABLE 5 (Continued)

	Cells		Matter		Earth	
	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2
Asian	0.64** (0.21)	0.54* (0.22)	0.58** (0.20)	0.4* (0.20)	0.77** (0.23)	0.5* (0.23)
Other	−0.4 (0.37)	0.26 (0.34)	0.05 (0.38)	−0.81* (0.31)	0.22 (0.42)	−0.1 (0.36)
Disadvantage	−0.3* (0.15)	0.05 (0.10)	NA	−0.19* (0.10)	−0.08 (0.10)	−0.27 (0.18)
Disability	0.04 (0.14)	0.15 (0.15)	0.24 (0.15)	−0.1 (0.15)	0.15 (0.17)	−0.17 (0.18)
ELL	0.13 (0.13)	0.36 (0.23)	−0.09 (0.22)	0.28 (0.23)	0.37 (0.25)	0.11 (0.27)
<i>Variance decomposition</i>						
School-level predictors	1.52 (0.33)	2.24 (0.50)	1.41 (0.34)	2.39 (0.51)	2.04 (0.47)	2.57 (0.65)
Classroom-level	0.44 (0.10)	0.44 (0.11)	0.64 (0.12)	0.23 (0.08)	0.79 (0.15)	0.49 (0.13)
Student-level	6.68 (0.15)	6.83 (0.16)	7.02 (0.16)	6.63 (0.16)	7.95 (0.19)	7.63 (0.19)

Note: Each column represents results for a model.

Estimates for random assignment blocks (school location) are omitted to save space.

* $p < .05$. ** $p < .01$. *** $p < .001$.

5 | RESULTS

5.1 | Impacts on teacher content knowledge and teacher instructional practice

Because the content group PD focused on improving teachers' content knowledge, we expected teachers in the Content group to have the highest scores. We treated the Content group as the reference in the models and compared the other two groups against it. The difference was only significant for the comparison between Content and control in the Earth unit (in a 25-question test, Content teachers on average answered three more questions correctly). It indicates that the Content intervention improved teachers' knowledge in the Earth unit but failed to significantly boost teachers' content knowledge in the other units. None of the teacher characteristics were significantly related to knowledge scores. Table 3 presents the detailed model estimates.

It was expected that Cog-sci teachers would have the highest implementation scores due to the focus of the Cog-sci intervention. Results show that it was true for the implementation of case comparisons and spaced testing. For case comparisons, Cog-sci teachers scored higher than other teachers in Cells and Matter. They scored higher than Content teachers in Earth. For spaced testing, Cog-sci teachers scored higher than control in Matter and higher than Content in Earth. However, the differences between Cog-sci and control were not significant for the implementation of visualization. The only significant differences were between Cog-sci and Content teachers for Cells and Matter. To sum

up, the Cog-sci intervention successfully boosted teachers' interpretation and application of two cognitive science principles across multiple units. Detailed HLM estimates are included in Table 4.

5.2 | Impacts on student achievement

5.2.1 | Impacts on student end-of-unit tests

Would the implementation of cognitive science principles lead to better student learning for the Cog-sci group? Would the content intervention have an impact on student achievement through other channels although it did not significantly improve teacher knowledge? Table 5 shows the coefficient estimates based on the HLMs. Most of the estimates for Cog-sci are positive while most of the estimates for Content are negative. However, none of the comparisons against the control group is significant.

To help interpret the magnitude of the estimates, effect sizes (ESs) for pairwise comparisons were calculated using the estimated differences and the pooled standard deviations across treatment conditions. They are plotted in Figure 3. ESs (x-axis) were presented for all combinations of unit (shape and color of the dot), cohort (top and bottom panel), and comparison (left and right panel). Because Cog-sci had the highest estimates most of the time, we plotted Cog-sci versus Content and Cog-sci versus control. The ESs for Cog-sci versus control ranged from 0 to 0.20. The ESs for Cog-sci versus Content ranged from 0.06 to 0.36. Although some differences were marginally significant (Cohort 1 Cells and Cohort 2 Earth between Cog-sci and Content, Cohort 1 Matter between Cog-sci and control), the only significant difference occurred for Cohort 2 Matter between Cog-sci and Content.

5.2.2 | Impact on state assessment

For models with total state test scores as the response variable, results were similar to those from the end-of-unit test analyses. The coefficient estimates for Cog-sci were all positive, but the differences between Cog-sci and control were not significant. The results are shown in Table 6 and illustrated in Figure S2. In Figure S2, the estimate means are shown on the left panel while the effect sizes are displayed on the right.

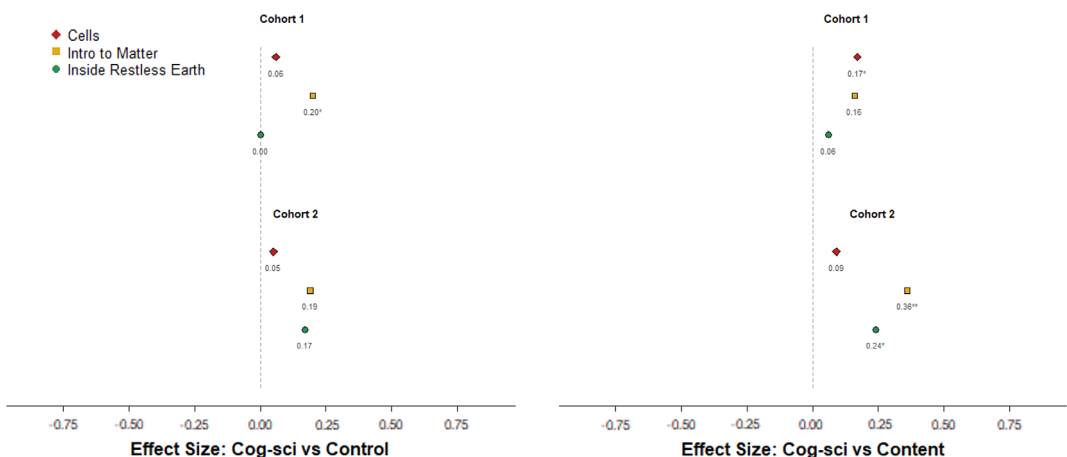


FIGURE 3 Effect sizes from main impact model [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 6 Coefficients estimates from HLMs, state test and subtest scores as response

	State test	Cells subtest		Matter subtest		Earth subtest	
	Both cohorts	Cohort 1	Cohort 2	Cohort 1	Cohort 2	Cohort 1	Cohort 2
<i>School-level predictors</i>							
Content intervention indicator	−1.02 (0.66)	−0.05 (0.10)	−0.35 (0.26)	−0.11 (0.12)	−0.11 (0.12)	−0.10 (0.24)	−0.22 (0.10)
Cog-sci intervention indicator	0.87 (0.66)	0.21 (0.10)	0.28 (0.25)	0.02 (0.12)	0.12 (0.11)	0.15 (0.24)	0.04 (0.10)
<i>Student-level predictors</i>							
Centered student fifth-grade math score	0.005*** (0.00)	0.001*** (0.00)	0.001* (0.00)	0.001* (0.00)	0.001** (0.00)	0.001** (0.00)	0.001** (0.00)
Centered student sixth-grade math score	0.009*** (0.00)	0.001*** (0.00)	0.002*** (0.00)	0.001*** (0.00)	0.001*** (0.00)	0.002*** (0.00)	0.001*** (0.00)
Centered student 5th-grade Reading score	0.013*** (0.00)	0.001** (0.00)	0.004*** (0.00)	0.001*** (0.00)	0.001*** (0.00)	0.003*** (0.00)	0.001*** (0.00)
Centered student sixth-grade Reading score	0.018*** (0.00)	0.002*** (0.00)	0.005*** (0.00)	0.002*** (0.00)	0.002*** (0.00)	0.005*** (0.00)	0.001*** (0.00)
Female	−1.48*** (0.17)	−0.05 (0.04)	−0.21* (0.08)	0.09 (0.05)	−0.17*** (0.05)	−0.16* (0.08)	−0.06 (0.04)
Black	−1.17*** (0.32)	−0.14 (0.07)	−0.54*** (0.15)	0.08 (0.08)	−0.16 (0.08)	−0.29* (0.14)	−0.21** (0.06)
Hispanic	−0.75* (0.35)	0 (0.08)	−0.3 (0.17)	0.01 (0.09)	−0.21* (0.09)	−0.26 (0.16)	−0.1 (0.07)
Asian	2.45*** (0.42)	0.36*** (0.10)	0.51* (0.20)	0.55*** (0.11)	0.22* (0.11)	0.72*** (0.19)	0.17 (0.08)
Other	0.76 (0.69)	0.45* (0.18)	−0.09 (0.32)	0.28 (0.20)	−0.21 (0.17)	0.48 (0.34)	0.06 (0.13)
Disadvantage	−0.68** (0.22)	−0.14 (0.07)	−0.21* (0.09)	−0.11 (0.08)	−0.01 (0.05)	−0.28* (0.14)	−0.09* (0.04)
Disability	0.38 (0.28)	0.30*** (0.06)	0.01 (0.14)	−0.14 (0.07)	−0.11 (0.07)	0.13 (0.12)	−0.07 (0.06)
ELL	0.66* (0.31)	0.00 (0.07)	−0.07 (0.20)	0.09 (0.07)	−0.02 (0.11)	0.15 (0.13)	−0.06 (0.08)
<i>Variance decomposition</i>							
School-level predictors	5.15 (1.00)	0.09 (0.02)	0.59 (0.14)	0.14 (0.03)	0.11 (0.03)	0.62 (0.13)	0.08 (0.02)
Student level	60.95 (0.94)	1.65 (0.04)	7.27 (0.16)	2.05 (0.04)	2.05 (0.04)	6.06 (0.13)	1.29 (0.03)

Note: Each column represents results for a model.

Estimates for random assignment blocks (school location) are omitted to save space.

* $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 7 Coefficients estimates from HLMs, Cohort 2 students that were taught by returning teachers

	Cohort 2 students that were taught by returning teachers		
	Cells	Matter	Earth
<i>School-level predictors</i>			
Content intervention indicator	−0.36 (0.60)	−0.36 (0.60)	−0.35 (0.87)
Cog-sci intervention indicator	0.5 (0.64)	0.5 (0.64)	0.87 (0.82)
Centered school Avg fifth-grade math score	−0.01 (0.01)	−0.01 (0.01)	−0.01 (0.01)
Centered school Avg sixth-grade math score	0 (0.01)	0 (0.01)	0 (0.01)
Centered school Avg fifth-grade Reading score	0.02* (0.01)	0.02* (0.01)	0 (0.01)
Centered school Avg sixth-grade Reading score	0.002 (0.01)	0.002 (0.01)	0.01 (0.01)
<i>Classroom-level predictors</i>			
Centered classroom Avg 5th-grade math score	0.004 0.00	0.004 0.00	0.007 (0.01)
Centered classroom Avg sixth-grade math score	0.01 0.00	0.01 0.00	0 (0.01)
Centered classroom Avg fifth-grade Reading score	0 0.00	0 0.00	0 0.00
Centered classroom Avg sixth-grade Reading score	0 0.00	0 0.00	0.01 0.00
<i>Student-level predictors</i>			
Centered student fifth-grade math score	0.002*** 0.00	0.002*** 0.00	0.0005*** 0.00
Centered student sixth-grade math score	0.002*** 0.00	0.002*** 0.00	0.003*** 0.00
Centered student fifth-grade Reading score	0.003*** 0.00	0.003*** 0.00	0.003*** 0.00
Centered student sixth-grade Reading score	0.004*** 0.00	0.004*** 0.00	0.004 0.00
Female	−0.30** (0.11)	−0.30** (0.11)	−0.77*** (0.12)
Black	−0.46* (0.20)	−0.46* (0.20)	−0.58** (0.20)

(Continues)

TABLE 7 (Continued)

	Cohort 2 students that were taught by returning teachers		
	Cells	Matter	Earth
Hispanic	−0.25 (0.22)	−0.25 (0.22)	−0.18 (0.22)
Asian	0.59* (0.25)	0.59* (0.25)	0.52* (0.26)
Other	−0.06 (0.39)	−0.06 (0.39)	0.13 (0.41)
Disadvantage	0.07 (0.11)	0.07 (0.11)	−0.29 (0.20)
Disability	0.08 (0.17)	0.08 (0.17)	−0.35 (0.22)
ELL	0.38 (0.26)	0.38 (0.26)	0.05 (0.34)
<i>Variance decomposition</i>			
School-level predictors	1.52 (0.33)	1.52 (0.33)	2.57 (0.65)
Classroom level	0.44 (0.10)	0.44 (0.10)	0.49 (0.13)
Student level	6.68 (0.15)	6.68 (0.15)	7.63 (0.19)

Note: Each column represents results for a model.

Estimates for random assignment blocks (school location) are omitted here to save space.

* $p < .05$. ** $p < .01$. *** $p < .001$.

As mentioned in the Measures section, we also constructed aligned subtests from the state test and used the subtest scores as additional outcome measures. The results are also shown in Table 6. Similar to the student end-of-unit test results, estimates for Cog-sci were positive but the differences were not significant. The comparison between Cog-sci and control was only marginally significant for the Cells Cohort 1 subtest. The comparisons between Cog-sci and Content were significant for both cohorts of the Cells subtest and Cohort 2 of the Earth subtest.

5.3 | Analyses on teaching for a second time

To test if prior year's experiences receiving the intervention help Cog-sci teachers more in promoting student learning in the second year, we restricted the Cohort 2 student sample to those who were taught by returning teachers and evaluated whether or not the treatment effects were larger. Table 7 and Figure S3 summarize the mean estimates for each of the treatment arms in Year 2, when only students taught by second-time teachers were included in the analyses. We did observe that the effects

TABLE 8 HLM estimates from models testing moderating effects of teaching experience teaching other units, Cohort 1 earth

	Earth
	Cohort 1 teacher experience
<i>Interactions</i>	
Content by experience	0.51 (0.87)
Cog-sci by experience	2.43** (0.72)
<i>School-level predictors</i>	
Content intervention indicator	−1.02 (0.65)
Cog-sci intervention indicator	−1.7** (0.61)
<i>Student-level predictors</i>	
Experience/exposure	−0.74 (0.51)
Centered student fifth-grade math score	0.002*** 0.00
Centered student sixth-grade math score	0.001** 0.00
Centered student fifth-grade Reading score	0.001*** 0.00
Centered student sixth-grade Reading score	0.004*** 0.00
Female	−0.09 (0.10)
Black	−0.43* (0.18)
Hispanic	−0.31 (0.19)
Asian	0.87*** (0.23)
Other	0.25 (0.42)
<i>Variance decomposition</i>	
School-level predictors	1.38 (0.41)

(Continues)

TABLE 8 (Continued)

	Earth
	Cohort 1 teacher experience
Classroom level	1.61 (0.65)
Student level	7.95 (0.19)

Note: Estimates for random assignment blocks (school location) are omitted here to save space.

* $p < .05$. ** $p < .01$. *** $p < .001$.

sizes for Cog-sci against other groups were larger, although not significant, probably due to sample size.

If Cog-sci teachers' prior experiences teaching the same unit might benefit their students, would Cog-sci teachers' prior experiences teaching other study-targeted units make a difference? We asked this question because the underlying principles of the Cog-sci intervention were consistent across units and teaching other units would also give teachers the opportunity to practice those principles. To test this assumption, we created a variable indicating how many other units the teacher had taught before teaching the unit under study. For Cohort 1 Matter, the indicator took a value of 0 or 1. For Cohort 1 Earth, the indicator ranged from 0 to 3 (Cohort 2 Cells was always taught before Cohort 1 Earth). For Cohort 2 Cells, Matter, and Earth, the indicator varied from 0 to 1, 0 to 4, and 0 to 5, respectively.

The main effect of the indicator, as well as the interaction terms between treatment and the indicator, was added to the main impact analysis model. Results show that both interaction terms for Cohort 1 Earth were significant, meaning that prior teaching experiences with Cells and Matter helped Cog-sci teachers significantly more than they helped control or Content teachers. The results indicate that in some circumstances accumulative experience with the Cog-sci intervention helped student learning. The interaction effect is shown in Table 8 and is illustrated in Figure S4.⁵ The y-axis represents the estimated average end-of-unit scores and the x-axis represents whether the teacher taught one or two prior units. The treatment groups are distinguished by the style of the lines. No significant interactions were observed from other models.

6 | DISCUSSION

Every year, millions of dollars are invested in teacher PD, and requirements are put into place to ensure that teachers engage in PD activities (Gore et al., 2017). This study responds to the call for the use of randomized designs in evaluating teacher PD (Wayne, Yoon, Zhu, Cronen, & Garet, 2008) and contributes to the literature on scaling up educational interventions (McDonald, Keesler, Kauffman, & Schneider, 2006). Despite the fact that most cognitive science research is lab based and that applying it to educational practice is extremely challenging, our study showed that curriculum modifications can be implemented by regular middle school classroom teachers with modest amount of PD, even in an urban-school setting. Although most of the effects on student achievement were not significant, it is encouraging that the Cog-sci intervention was able to change teacher practice and moved student achievement in the right direction.

6.1 | Implications

In general, we believe our findings support the view that there are links between professional learning in science, changes in practice, and student learning (e.g., Heller, Daehler, Wong, Shinohara, & Miratrix, 2012). However, the moderate relationships and nonsignificant effects on learning suggest several hypotheses. Fullan and Miles (1992) mentioned that while implementing new materials, there might be a stage called implementation dip, when things get worse before they get better. And some studies have shown that it might take longer for new interventions to have an effect on student achievement (Garet et al., 2001). This might be the case for the current study.

Pine et al. (2006) mentioned that while some students may learn more from a new curriculum that encourages more direct engagement, others may not benefit at all. A majority of students participating in the current study scored well below the state average on the state science test, and the participating district ranked among the lowest 10% in the state in student achievement. As some participating teachers indicated, the modified material of the Cog-sci intervention was too detailed for some students and might have confused rather than helped them. It would be useful to modify the materials so that students at different levels can understand them with some guidance and so that teachers can easily integrate them into their instruction with different groups of students. Moreover, consistent with Evans (1996), the degree of student mobility and absences from school also had a negative impact on their learning. Many students did not receive the full “dose” of the intervention, yet took the end-of-unit test. Related to these issues were the generally low levels of reading readiness of students, which may have rendered moot even the most carefully constructed cognitive science-based lesson (Akbasli, Sahin, & Yaykiran, 2016).

As previously shown by similar studies, accountability pressures faced by schools may lead to low levels of implementation of curricula (Harris et al., 2015). Some teachers mentioned that they were not able to finish the unit and had to shift focus to test preparation toward the end of the semester, especially in the case of the Earth unit, which was taught in the spring. As suggested by Penuel et al. (2011), a more flexible, loosely formatted curriculum guide was more likely to be successful when teachers experienced conflicting messages about their teaching and lesson plans.

We should also consider the organizational factors that influence PD effectiveness (van Driel et al., 2012). The study was carried out in a large urban school district, and teacher attrition was a concern. After the first year of the intervention, approximately 40% of the teachers were no longer teaching science in the same grade or in the same school due to various reasons, none related to the study itself. While we provided makeup days for the PD and stipends for newly joined teachers, these were stopgap measures. Another limitation was instability in school leadership and district leadership. Harris et al. (2015) indicated that the most important factor for their study's positive findings was the alignment between the new curriculum and administrative support. In a large, diverse urban district, efforts to promote implementation must cohere with other elements of the local system (Harris et al., 2015). A previous administrative commitment to science teaching could be undone by a new principal or a new superintendent with different priorities. As a result, teachers may only implement the new materials to a limited degree once they perceive the learning goals to be incongruent with the changing district standards (Penuel & Gallagher, 2009). As with new teachers, new principals might have the effect of lessening the duration and potency of the interventions (Boruch, Merlino, & Porter, 2012). These are common issues in urban education in the United States, and designers of new interventions must plan ahead for the changing leadership environment and priorities. Teacher content knowledge was believed to be key in promoting student achievement (Diamond et al., 2014). However, results for the Content intervention did not show promise, at least for the amount of PD that was provided. Teacher content knowledge failed to improve, and it did not

translate into improved student achievement. The results contribute to a growing body of evidence (e.g., *Garet et al., 2001*) demonstrating that improving teacher content knowledge through professional development is a difficult task. Our results provide no evidence that the Content PD boosted teachers' content knowledge and, not surprisingly, no evidence of benefits to student achievement, similar to *Roblin, Schunn, and McKenney (2018)*.

One possible explanation why the Cog-sci intervention but not the Content intervention had some effects might be the deliberate exclusion of discussions of pedagogical content knowledge (PCK) and student learning—which was intended to maintain clear distinctions between the two treatment arms but which may have made the content of the sessions more inert for teachers in the Content-only arm (*Schneider & Plasman, 2011*). This interpretation is consistent with previous research emphasizing the importance of PCK. PCK is theorized to be multifaceted, incorporating aspects of knowledge such as knowing appropriate models to illustrate concepts and understanding how students learn in a particular content area (*Shulman, 1986*). As emphasized by *Harris et al. (2015)*, no matter how groundbreaking and well-designed the new curriculum is, teacher PD should instruct the teachers in how to integrate and adapt the materials to fit their teaching goals. There is empirical evidence that a profound understanding of instructional content helps teachers to instruct in ways that support student learning (*Ball, 1991, 2000; Ball, Lubienski, & Mewborn, 2001; Ball, Thames, & Phelps, 2008*). While the Cog-sci arm did address PCK more than did the Content arm, it did not go all the way from introducing cognitive science principals to explicating for teachers how to transform their instruction by using the new materials. As prior studies have found, just embedding supports into materials is insufficient and could limit the efficacy of the curriculum materials (*Borman et al., 2008; Penuel et al., 2011*), and teachers often find it difficult to discern strategies for engaging students in scientific inquiries from materials alone (*Alozie, Moje, & Krajcik, 2010*). The Cog-sci teachers might have benefited more if our PD had offered more direct experiences with the optimal learning environment they are expected to construct (*Marx & Harris, 2006*).

Another explanation is that average effects may be masking substantial individual variation. Previous research has shown that what teachers learn in PD depends largely on the existing knowledge they bring to the activity and that they can have quite different takeaways from their learning experiences (*Minor, Desimone, Lee, & Hochberg, 2016*). Personalizing PD to address teachers' particular circumstances, knowledge, and experience holds promise for increasing their effectiveness (*Starkey et al., 2009*).

6.2 | Limitations

In interpreting the results of our study, several limitations should be considered. Teacher practice measures were constructed based on survey measures, which might not truly capture changes in the classroom (*Desimone, 2009*). Using and coding videos can yield more reliable information about instructional practices (*Lee et al., 2016*) but would be too costly in this case. Also, with a sample of 90 schools, we had to rely on random assignment for baseline equivalence between teachers in different arms due to the lack of pretest of teacher knowledge. In many schools, grade assignment for teachers was not known until only a couple days prior to the start of teacher PD, so we could not administer a pretest of teacher knowledge to teachers before PD. Due to time limits, we did not have the luxury to administer it on the first day of training either. Moreover, without a premeasure of key mediating variables, the authors were handicapped in identifying the teachers who benefited the most or the least from the PD. In the future, we would try to extend the training hours and have the pretest

scheduled at the beginning of the PD session so this measure could be used in evaluating the impact of PD on teacher knowledge.

7 | CONCLUSIONS

Nevertheless, the above discussion of our work should not be taken in a way that obfuscates the strengths of the work. First, both the Cog-sci professional development and materials intervention and the Content intervention were serious efforts designed to be consistent with what is known about effective teacher professional development (Desimone, 2009; Garet et al., 2001). Second, the interventions were tested using the gold standard: A large randomized control trial that maintained its integrity through to the end of the study. Third, highly aligned and reliable outcome measures were created and used to operationalize teacher content knowledge and student achievement on the goals of the science interventions. The effort is far from the practice of a one-day workshop evaluated with a quasi-experimental design using distant and unaligned state achievement tests.

We believe our study provides valuable insights for future teacher PD that meets the ambitious goals embodied in the NGSS (NGSS Lead States, 2013) as well as insights for curriculum studies because course materials serve as a vehicle of reform by supporting the enactment of specific instructional practices (Remillard, Harris, & Agodini, 2014). Our study supports the view that PD grounded in pedagogical content knowledge is more effective than PD focused on content only. Our findings also suggest the importance of PD materials and strategies that are aligned with district goals and priorities and are calibrated to individual teacher and classroom needs, in contrast to the “one size fits all” method of PD (Desimone & Garet, 2015), and they suggest the considerable influence that organizational factors, such as resources, time, and mobility, are likely to have on the outcomes of PD (van Driel et al., 2012).

The study sheds light on the importance of designing an intervention that can be effective in school districts that are experiencing a high magnitude of teacher instability and are under pressure to meet accountability requirements (Harris et al., 2015; Penuel, Riel, Krause, & Frank, 2009). Such an intervention will be a challenge for researchers and practitioners, but it appears to bring benefits to students, especially those in high-poverty areas. At the same time, as science curricula and instructional practices and supporting infrastructure for science education continue to evolve in the United States, one reasonable strategy would be to apply cognitive science principles to new instructional formats, such as multi-year curriculum sequences, learning activities that engage a broader array of science practices, and technology-enabled learning, and methodically evaluate the effects.

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ENDNOTES

¹ Because we already knew whether a teacher participated in the study for Cohort 2 students, we included the interactions between that indicator and the treatment variables in the model to predict Cohort 1 students' test scores. All three models (for the three units) showed nonsignificant interactions, indicating that teachers' attrition would not lead to discrepancies in student achievement across the three arms.

² We decided to use a one-parameter model because we want all the items to contribute equally to the ability estimates. We also tested a two-parameter model, and it showed that the slope coefficients were not very different across items.

- ³ The alignment indices ranged from 0.24 to 0.32, which were moderate. The alphas were higher than 0.7 except for two subtests, where the low numbers (0.47 and 0.54) were due to the small number of items in the subtests.
- ⁴ We also fit four-level HLMs where there was an additional teacher level. Results were not very different, and the same conclusions could be drawn based on those models.
- ⁵ All Earth teachers had taught at least Matter previously. Therefore, the indicator only took a value of 1 or 2. It was actually an indicator for whether the teacher taught Cells previously.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

How to cite this article: Yang R, Porter AC, Massey CM, Merlino JF, Desimone LM. Curriculum-based teacher professional development in middle school science: A comparison of training focused on cognitive science principles versus content knowledge. *J Res Sci Teach*. 2019;1–31. <https://doi.org/10.1002/tea.21605>